

Rajalakshmi Engineering College

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Department: AI & ML - Section 4
Batch: 2028
Degree: B.E - AI & ML

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 1_Q1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Gloria is responsible for monitoring the performance of two machines in a factory. She needs to determine which of the two machines is operating closest to the optimal temperature of 100 degrees Celsius using the relational operator.

Assist Gloria in displaying the machine's temperature, which is closer to 100, and the difference from 100.

Input Format

The first line of input consists of an integer N, representing the temperature of the first machine.

The second line consists of an integer M, representing the temperature of the second machine.

Output Format

The output prints "The integer closer to 100 is X with a difference of Y" where X is the temperature of the closer machine and Y is the difference from 100.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 90

80

Output: The integer closer to 100 is 90 with a difference of 10

Answer

```
// You are using Java
import java.util.Scanner;
```

```
public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        int N = scanner.nextInt(); // Temperature of first machine
        int M = scanner.nextInt(); // Temperature of second machine

        int diffN = Math.abs(N - 100);
        int diffM = Math.abs(M - 100);

        int closer = (diffN < diffM) ? N : M;
        int difference = (diffN < diffM) ? diffN : diffM;

        System.out.print("The integer closer to 100 is " + closer + " with a difference
of " + difference);

        scanner.close();
    }
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 1_Q2

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. PROBLEM STATEMENT:

Dave got two students who want help with their doubt. Each hands out an integer and wants to find if one integer is positive while the other is not divisible by 3. Write a program to achieve this and conclude for them.

Input Format

The first line of input represents the first integer.

The second line of input represents the second integer.

Output Format

The output should display as "One of the integers is positive while the other is not divisible by 3." or "Neither of the integers meets the condition."

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4

3

Output: One of the integers is positive while the other is not divisible by 3.

Answer

```
// You are using Java
import java.util.Scanner;
```

```
public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        int first = scanner.nextInt();
        int second = scanner.nextInt();

        boolean firstPositive = first > 0;
        boolean secondPositive = second > 0;

        boolean firstNotDivBy3 = first % 3 != 0;
        boolean secondNotDivBy3 = second % 3 != 0;

        if ((firstPositive && secondNotDivBy3) || (secondPositive && firstNotDivBy3))
        {
            System.out.print("One of the integers is positive while the other is not
divisible by 3.");
        } else {
            System.out.print("Neither of the integers meets the condition.");
        }

        scanner.close();
    }
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 2_Q3

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

John is a fitness trainer, and he wants to use the BMI calculator to assess the body mass index of his clients. He has a list of clients based on their height and weight.

John plans to write a program to quickly determine the BMI and provide a classification for each client.

If BMI is less than 18.5, the program will classify it as "Underweight" If BMI is between 18.6 and 24.9, the program will classify it as "Normal Weight" If BMI is between 25.0 and 29.9, the program will classify it as "Overweight" If BMI is 30.0 or higher, the program will classify it as "Obese"

Note: Formula to calculate BMI = $\text{weight}/(\text{height}*\text{height})$

Input Format

The first line of input consists of a double value, representing the height of the person in meters.

The second line consists of a double value, representing the weight of the person in kilograms.

Output Format

The first line of output prints "BMI: " followed by a double (rounded to two decimal places) representing the calculated BMI.

The second line prints "Classification: " followed by a string indicating the BMI category (Underweight, Normal Weight, Overweight, or Obese).

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1.2

45.2

Output: BMI: 31.39

Classification: Obese

Answer

```
// You are using Java
import java.util.Scanner;
```

```
public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
```

```
        double height = scanner.nextDouble();
        double weight = scanner.nextDouble();
```

```
        double bmi = weight / (height * height);
        String classification;
```

```
        if (bmi < 18.5) {
            classification = "Underweight";
        } else if (bmi >= 18.6 && bmi <= 24.9) {
            classification = "Normal Weight";
```

```
} else if (bmi >= 25.0 && bmi <= 29.9) {  
    classification = "Overweight";  
} else {  
    classification = "Obese";  
}  
  
System.out.printf("BMI: %.2f%n", bmi);  
System.out.print("Classification: " + classification);  
  
scanner.close();  
}  
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 5_Q5

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Ram is working as a developer for BrightEdu Coaching Center, which wants to build a student fee management system.

Each student's enrollment has:

An Enrollment ID (integer) A Student Name (string) The Number of Subjects (integer)

The fee calculation rules are:

Registration Fee = 1000 units (flat for every student). Per Subject Fee = 800 units. If the student enrolls in more than 5 subjects, a 20% scholarship (discount) is applied on the total fee.

Ram has been asked to implement this system using:

A class with attributes for student details. A constructor to initialize student details. Setter methods to update details if needed. Getter methods to retrieve details. Objects of the class to represent student enrollments.

Finally, display each student's details and final fee.

Input Format

The first line of input contains an integer N, representing the number of students.

For each student:

- The next line contains the Enrollment ID (integer).
- The following line contains the student's name (string).
- The next line contains the Number of subjects (integer).

Output Format

For each student, print the details in the following format:

- Enrollment ID: <enrollment_id>
- Student Name: <student_name>
- Final Fee: <final_fee> (rounded to one decimal place)

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

1234

Ravi Kumar

3

Output: Enrollment ID: 1234

Student Name: Ravi Kumar

Final Fee: 3400.0

Answer

```
// You are using Java
import java.util.Scanner;
```

```
class Student {
```

```
private int enrollmentId;  
private String studentName;  
private int numberOfSubjects;
```

```
public Student(int enrollmentId, String studentName, int numberOfSubjects) {  
    this.enrollmentId = enrollmentId;  
    this.studentName = studentName;  
    this.numberOfSubjects = numberOfSubjects;  
}
```

```
public int getEnrollmentId() {  
    return enrollmentId;  
}
```

```
public String getStudentName() {  
    return studentName;  
}
```

```
public int getNumberOfSubjects() {  
    return numberOfSubjects;  
}
```

```
public double calculateFinalFee() {  
    double registrationFee = 1000;  
    double subjectFee = numberOfSubjects * 800;  
    double totalFee = registrationFee + subjectFee;  
  
    if (numberOfSubjects > 5) {  
        totalFee = totalFee - (totalFee * 0.20);  
    }  
  
    return totalFee;  
}
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        int N = Integer.parseInt(sc.nextLine());
```

```
for (int i = 0; i < N; i++) {  
    int enrollmentId = Integer.parseInt(sc.nextLine());  
    String studentName = sc.nextLine();  
    int numberOfSubjects = Integer.parseInt(sc.nextLine());  
  
    Student student = new Student(enrollmentId, studentName,  
numberOfSubjects);  
    double finalFee = student.calculateFinalFee();  
  
    System.out.printf("Enrollment ID: %d\n", student.getEnrollmentId());  
    System.out.printf("Student Name: %s\n", student.getStudentName());  
    System.out.printf("Final Fee: %.1f\n", finalFee);  
}  
    sc.close();  
}  
}
```

Status : Correct

Marks : 10/10